

Agents and Environments

Reinforcement learning studies an agent that interacts with an environment over discrete time steps. At each step the agent observes a state, chooses an action, and receives a scalar reward together with the next state.

The agent's goal is to maximise the expected cumulative reward, not the immediate reward. Future rewards are usually discounted by a factor between 0 and 1, so that rewards further in the future count less.

Value Functions

The state value function gives the expected return when starting from a state and following a given policy thereafter. The action value function, often written Q , gives the expected return when taking a specific action in a state and following the policy afterwards.

The Bellman equation expresses the value of a state in terms of the values of its successor states, which is what makes dynamic programming and temporal difference learning possible.

Exploration and Exploitation

An agent that always takes the action it currently believes is best may never discover a better one. An agent that always explores randomly never accumulates reward. Balancing the two is the exploration-exploitation trade-off.

The epsilon-greedy strategy is the simplest practical answer: with probability epsilon take a random action, otherwise take the action with the highest estimated value. Epsilon is usually decayed over training.

Q-Learning

Q-learning is an off-policy temporal difference method that updates the action value estimate towards the reward plus the discounted maximum value of the next state. Being off-policy means it learns about the greedy policy while behaving according to a different, more exploratory policy.

Q-learning converges to the optimal action value function under the conditions that every state-action pair is visited infinitely often and the learning rate decays appropriately.